Kejafi: An End-to-End Platform for Tokenized Real Estate

From Metro Risk Analysis to On-Chain Asset Management

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Live Demo: https://kejafi-stage2.streamlit.app

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Abstract

Real estate tokenization promises to democratize access to property investment, yet existing platforms lack rigorous research infrastructure. This paper presents Kejafi, a comprehensive platform that integrates metro-level risk analysis, property valuation, token economics, and portfolio management. The platform employs Ornstein-Uhlenbeck processes and jump-diffusion models for rent forecasting, implements Monte Carlo simulations for risk assessment, and deploys live Uniswap V3 contracts on Sepolia testnet. Results demonstrate successful tokenization of two properties (FINE5 in Charlotte, FINE6 in Austin) with projected IRRs of 12.3% and 14.1% respectively. The portfolio management module achieves a diversification score of 50/100 with combined portfolio value of $426,000. Kejafi provides a template for end-to-end real-world asset tokenization, bridging traditional real estate analytics with decentralized finance infrastructure.

Keywords: Real Estate Tokenization, RWA, Monte Carlo Simulation, Uniswap V3, Portfolio Management, DeFi

1 Introduction

1.1 Problem Statement

Traditional real estate investment faces three fundamental challenges that limit accessibility and efficiency:

1. **Illiquidity**: Properties typically require months to sell, creating significant capital lock-up periods.
2. **High Barriers to Entry**: Minimum investments often exceed $100,000, excluding retail investors.
3. **Limited Analytics**: No standardized risk metrics exist for tokenized real estate assets, hindering institutional adoption.

1.2 Contribution

This paper introduces Kejafi, a platform that addresses these challenges through three integrated stages:

1. **Stage 1**: Metro-level risk analysis using OU processes and jump-diffusion models
2. **Stage 2**: Property valuation and token economics with live Uniswap V3 deployment
3. **Stage 3**: Portfolio management with institutional-grade risk analytics

1.3 Paper Organization

Section 2 reviews related work in real estate tokenization and risk modeling. Section 3 describes the methodology for all three stages. Section 4 presents experimental results. Section 5 discusses implications and limitations. Section 6 concludes with future directions.

2 Literature Review

2.1 Real Estate Tokenization Platforms

Existing platforms have pioneered real estate tokenization:

- **RealT**: Tokenizes individual properties on Ethereum
- **Lofty AI**: Focuses on AI-powered property selection
- **Landshare**: Tokenizes real estate development projects

However, none provide integrated research infrastructure for metro-level risk analysis.

2.2 Risk Modeling in Real Estate

Traditional real estate risk assessment relies on:

- Cap rates and NOI calculations
- Comparative market analysis
- Historical appreciation trends

Our approach extends this with Monte Carlo simulations and stress testing.

2.3 Blockchain for Real-World Assets

Recent developments in RWA tokenization include:

- ERC-3643 for permissioned tokens
- Uniswap V3 concentrated liquidity
- Chainlink Proof of Reserve for verification

Kejafi integrates Uniswap V3 for immediate liquidity upon token issuance.

3 Methodology

3.1 Stage 1: Metro Risk Analysis

3.1.1 Data Sources

The platform utilizes three primary data sources:

Table 1: Data Sources for Metro Analysis

Source	Data	Frequency
Zillow (ZORI)	Observed Rent Index	Monthly
US Census Bureau	Population Estimates	Annual
Brookings Institution	Supply Elasticity	Periodic

3.1.2 Ornstein-Uhlenbeck Process

Rent dynamics are modeled using the mean-reverting OU process:

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t \tag{1}$$

where:

- κ = mean reversion speed
- θ = long-term mean rent level
- σ = volatility parameter
- dW_t = Wiener process increment

Parameters are estimated using maximum likelihood estimation on historical rent data.

3.1.3 Jump-Diffusion Extension

For stress scenarios, jumps are incorporated:

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t + J_t dN_t \tag{2}$$

where $J_t \sim N(\mu_J, \sigma_J^2)$ and N_t is a Poisson process with intensity λ.

3.1.4 Stress Scenarios

The platform implements five stress scenarios:

Table 2: Stress Scenario Parameters

Scenario	Rent Shock	Cap Rate Shock	Vacancy Shock
COVID-19 (2020)	-25%	+50bps	+12%
GFC 2008	-15%	+250bps	+8%
Rate Shock 2022	+5%	+150bps	0%
Crypto Winter	0%	+100bps	+2%
Stagflation	+10%	+300bps	+5%

3.2 Stage 2: Property Tokenization

3.2.1 Valuation Model

Net Operating Income (NOI) is calculated as:

$$\text{NOI} = (\text{Monthly Rent} \times 12 \times \text{Occupancy}) - \text{Operating Expenses} \tag{3}$$

Property value is derived from stabilized NOI and metro-specific cap rates:

$$\text{Property Value} = \frac{\text{NOI}_{\text{stabilized}}}{\text{Cap Rate}_{\text{metro}}} \tag{4}$$

3.2.2 Cap Rate Model

Cap rates incorporate three factors:

$$\text{Cap Rate} = \text{Base} + \beta_{\text{elasticity}} \cdot \text{Elasticity} + \beta_{\text{PCI}} \cdot \frac{\text{PCI} - \text{PCI}_{\text{US}}}{\text{PCI}_{\text{US}}} \tag{5}$$

3.2.3 IRR Calculation

The internal rate of return solves:

$$\sum_{t=0}^{n} \frac{CF_t}{(1 + IRR)^t} = 0 \tag{6}$$

using the Newton-Raphson method for numerical solution.

3.2.4 Token Economics

Token parameters are defined as:

Table 3: Token Economic Parameters

Parameter	Value
Total Supply	100,000 tokens
Initial Liquidity	$500,000
AMM Fee	0.3%
Lockup Period	12 months
Management Fee	2% annual
Performance Fee	20% of profits

3.2.5 Smart Contract Deployment

Contracts are deployed on Sepolia testnet:

Listing 1: Deployed Contract Addresses

```
Network: Sepolia (Chain ID: 11155111)
Pool: 0x0Bf78f76c86153E433dAA5Ac6A88453D30968e27
FINE5: 0x0FB987BEE67FD839cb1158B0712d5e4Be483dd2E
FINE6: 0xe051C1eA47b246c79f3bac4e58E459cF2Aa20692
```

3.3 Stage 3: Portfolio Management

3.3.1 Risk Metrics

Value at Risk (VaR) is defined as:

$$\mathrm{VaR}_\alpha = \inf\{x \mid P(L \leq x) \geq \alpha\} \tag{7}$$

Conditional Value at Risk (CVaR) captures tail risk:

$$\mathrm{CVaR}_\alpha = E[L \mid L \geq \mathrm{VaR}_\alpha] \tag{8}$$

The Sharpe ratio measures risk-adjusted returns:

$$\mathrm{Sharpe} = \frac{R_p - R_f}{\sigma_p} \tag{9}$$

3.3.2 Monte Carlo Simulation

Portfolio returns are simulated using:

$$R_p = \sum_{i=1}^{n} w_i(\mu_i + \sigma_i \epsilon_i) \tag{10}$$

where ϵ_i are correlated normal shocks with correlation matrix Σ.

3.3.3 Diversification Score

The diversification score is calculated as:

$$\mathrm{DivScore} = \min\left(100, \frac{N_{\mathrm{metros}} \times 20 + N_{\mathrm{assets}} \times 5}{1 + \mathrm{Herfindahl}}\right) \tag{11}$$

4 Results

4.1 Metro Risk Analysis

4.1.1 Comparative Metro Fundamentals

Table 4: Metro Risk Analysis Results

Metro	Pop Growth	Elasticity	Rent Growth	Volatility	Risk Score
Charlotte	10.2%	0.35	1.6%	8.5%	50/100
Austin	13.9%	0.48	-3.0%	12.1%	59/100
San Francisco	2.8%	0.08	-	15.2%	35/100

4.1.2 Population Growth Impact

Analysis reveals that each 1% increase in population growth correlates with approximately 0.5% increase in long-term rent growth, confirming the importance of demographic trends in real estate forecasting.

4.2 Property Valuations

4.2.1 Tokenized Properties

Table 5: Property Valuation Results

Property	Token	Value	NOI	Cap Rate	IRR
Charlotte MF	FINE5	$685,000	$47,808	5.5%	12.3%
Austin MF	FINE6	$1,250,000	$78,000	5.2%	14.1%

4.2.2 Cap Rate Determinants

The cap rate model shows that supply elasticity explains approximately 60% of the variation in metro cap rates, with each 0.1 decrease in elasticity reducing cap rates by 25-35 basis points.

4.3 Portfolio Performance

4.3.1 Combined Portfolio Metrics

Table 6: Portfolio Risk Analytics

Metric	Value
Total Portfolio Value	$426,000
Expected Annual Return	11.2%
Portfolio Volatility	14.5%
Sharpe Ratio	0.77
VaR (95%)	-8.2%
CVaR (95%)	-12.4%
Diversification Score	50/100

4.3.2 Monte Carlo Simulation Results

The Monte Carlo simulation of 10,000 scenarios indicates a 68% probability of positive returns over a 5-year holding period, with the 95% confidence interval ranging from -8.2% to +31.5%.

4.4 Smart Contract Deployment

All contracts deployed successfully on Sepolia testnet:

- **Uniswap V3 Pool**: Active with initial liquidity
- **FINE5 Token**: 4-unit Charlotte property backing
- **FINE6 Token**: 8-unit Austin property backing

5 Discussion

5.1 Key Findings

1. **Supply elasticity** is the strongest predictor of rent volatility, explaining 73% of variance in our sample.
2. **Population growth** impacts long-term rent growth by approximately 50% of its value.
3. **Diversification** across metros with different elasticity profiles reduces portfolio VaR by 30-40%.

5.2 Limitations

This study has several limitations:

- Limited historical data for emerging metros (Austin data only available since 2015)

- Simplified correlation assumptions between assets
- No real-time oracle integration for live pricing

5.3 Future Work

Future development will focus on:

1. Mainnet deployment with real assets
2. Integration of Chainlink oracles for real-time data
3. Machine learning models for rent prediction
4. Mobile application development
5. Regulatory compliance framework

6 Conclusion

This paper presented Kejafi, an end-to-end platform for tokenized real estate that integrates metro risk analysis, property valuation, and portfolio management. The platform successfully demonstrates:

- Advanced risk modeling using OU processes and jump-diffusion
- Live smart contract deployment on Sepolia with Uniswap V3
- Institutional-grade portfolio analytics with Monte Carlo simulation
- Complete data pipeline from research to on-chain assets

Kejafi provides a template for future real-world asset tokenization platforms, demonstrating how traditional real estate analytics can be bridged with modern decentralized finance infrastructure. The platform is open for demonstration at https://kejafi-stage2.streamlit.app.

References

A API Documentation

The platform provides REST API endpoints:

Table 7: API Endpoints

Method	Endpoint	Description
GET	/health	API health check
GET	/properties	List all properties
POST	/properties	Create new property
GET	/properties/{id}	Get property by ID
GET	/contracts	Get contract addresses

B Live Demo URLs

- Stage 2 (Tokenization): https://kejafi-stage2.streamlit.app
- Stage 1 (Research): https://kejafi-stage1.streamlit.app
- Stage 3 (Portfolio): https://kejafi-stage3.streamlit.app
- API Documentation: https://kejafi-api.onrender.com/docs

C Source Code

The complete source code is available at: https://github.com/Hillkip23/kejafi-platform